

0 9

Read the following passage.

Kidney cells produce a glycoprotein hormone called erythropoietin (EPO). An EPO molecule contains 165 amino acids and approximately 50% of its mass is carbohydrate.

EPO is transported in the blood and stimulates the bone marrow to produce red blood cells. In this way, enough red blood cells are produced to maintain the blood's oxygen-carrying capacity. 5

Some athletes choose to increase their blood EPO concentration by injecting synthetic EPO. This practice is called blood boosting and is banned in sport as a form of drug abuse. Athletics' authorities use a programme of drug testing to detect athletes who have injected EPO. In this programme, an ELISA test is performed on urine samples to measure the concentration of EPO in the athlete. 10

Two types of monoclonal antibody are used in this ELISA test:

- anti-human EPO antibody, prepared by injecting human EPO into mice
- anti-mouse antibody, prepared by injecting anti-human EPO antibody into goats. An enzyme is attached to the anti-mouse antibody. 15

Use the information in the passage and your own knowledge to answer the following questions.

0 9 . 1

Kidney cells produce a glycoprotein called erythropoietin (EPO) (line 1).

Identify **two** organelles in kidney cells that enable the production of EPO.

[1 mark]

1 _____

2 _____

0 9 . 2

Explain the biological advantage to athletes of injecting synthetic EPO (lines 7–8).

[2 marks]



0 9 . 3

Describe how mice injected with human EPO produce anti-human EPO antibody (line 14).

[3 marks]

0 9 . 4

Describe the roles of anti-human EPO antibody and anti-mouse antibody with enzyme attached (lines 14–16) in producing a positive result for EPO in the ELISA test.

[3 marks]

Role of anti-human EPO antibody _____

Role of anti-mouse antibody with enzyme attached _____

0 9 . 5

Some people object to using monoclonal antibodies in testing programmes. Use information in the passage to suggest why.

[1 mark]

END OF QUESTIONS**10**